NV23 image231924 c01 closeout: aligned NV, Ramsey T_2^*, and ^{13}C check

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Executive summary

The project objective was to find a magnetic-field-aligned NV in the image231924 spatial range, then obtain a well-supported Ramsey T_2^* and ^{13}C conclusion. The first candidate, `image231924_c01`, was trackable and passed the requested strong-pi pulsed ODMR alignment screen. Weak-pi pulsed ODMR refined the microwave center, and a corrected-center Ramsey repeat completed normally. The terminal conclusion is:

- Magnetic-field-aligned NV found: `image231924_c01`.
- Ramsey signal: present, with a repeatable oscillatory component near 1.76 MHz.
- T_2^*: supported scale about 4.0 μs from a Gaussian-envelope Ramsey fit; model-dependent range from exponential/stretched/Gaussian fits is approximately 3.2–4.0 μs. This is a scale estimate, not sub-100-ns precision.
- ^{13}C: no resolved nearby ^{13}C coupling is supported by the corrected-center Ramsey FFT. The data do not show a clean carrier plus symmetric det$\pm f_{^{13}\mathrm{C}}$ sideband pair; the lower target bin is embedded in a broad carrier/detuning-like band and the upper target bin is weak.

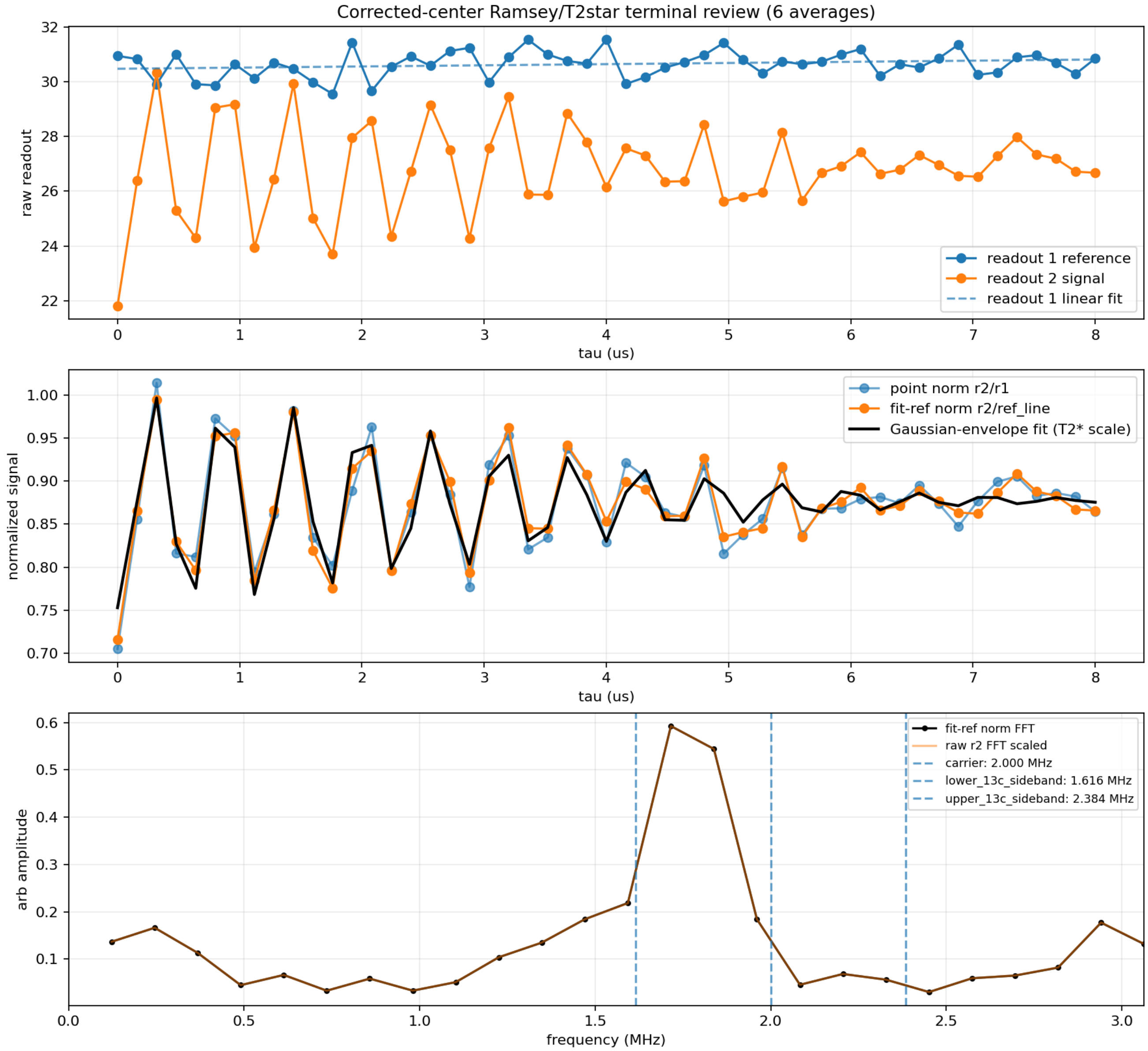
Evidence chain

Step	Result	Main evidence
Image export and seed	Explicit image-axis contract; c01 seed selected	candidate list JSON
TrackCenter	c01 passed, 36.008 kcps first pass	bridge result 23:36
Strong-pi pODMR retry	Visible resonance near 3.879 GHz, 23–25% normalized dip	analysis 00:38
Weak-pi pODMR	Usable center 3.8758666667 GHz, grid-limited	analysis 01:16
First Ramsey scout	Oscillation present; T2*/13C ambiguous	analysis 02:49
Narrow weak-pi refresh	Updated center 3.8761166667 GHz	analysis 03:17
Corrected-center Ramsey	Terminal T2*/13C closeout	analysis 05:18

Terminal Ramsey acquisition

The terminal corrected-center Ramsey job was the 03:21 EDT execute for c01. It ran `ramsey.xml` with $\tau = 0 \ldots 8$ μs, 51 points, 6 averages, and 100000 repetitions per average. The bridge completed normally at 04:51 EDT. Auto-align counts were 23.416 kcps and post-run final counts were 26.645 kcps. The saved experiment is the 2026-05-12 03:24:49 Ramsey savedexperiment.

The review used raw readouts plus two normalized views: point-wise r_2/r_1 and signal divided by a linear fit to readout 1. The sequence metadata identifies readout 1 as the pre-Ramsey $m_S = 0$ reference and readout 2 as the post-Ramsey signal for `full_experiment=0`.

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Terminal claim status:
- Ramsey signal: present near 1.76 MHz in raw r2 and normalized data.
- T2*: about 4.0 us (Gaussian envelope); model range ~3.2-4.0 us.
- 13C: no resolved coupling supported; no clean carrier +/- f13C sideband pair.
- FFT ranks: carrier 5, lower/upper 13C bins 3/19.
- Drift: scan-order-aware, flagged averages [].
- Next: optional only if a stronger null limit or higher-precision T2* is desired.
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Figure 1: Terminal corrected-center Ramsey review. The top panel shows raw readouts, the middle panel shows normalized signal and the Gaussian-envelope fit, and the bottom panel shows the FFT with planned carrier and ^{13}C sideband markers.

T_2^* assessment

The Ramsey oscillation is visible in raw readout 2 and in fitted-reference-normalized signal. The two largest FFT bins are 1.716 MHz and 1.838 MHz, with an amplitude-weighted center near 1.774 MHz. Stored-average support is moderate and consistently positive: 15 pairwise detrended normalized-trace correlations have median 0.571 and range 0.375–0.659.

The preferred empirical fit is a Gaussian-envelope Ramsey decay on the fitted-reference-normalized signal. It gives

$$T_2^* \approx 4.03\ \mu\text{s}, \qquad f \approx 1.759\ \text{MHz}.$$

An exponential-envelope fit gives $T \approx 3.16\ \mu$s, and a stretched fit gives $T \approx 3.89\ \mu$s with stretch exponent about 1.70. The project should therefore report a supported T_2^* scale of about 4 μs, with model-dependence at the few-tenths-to-one-microsecond level.

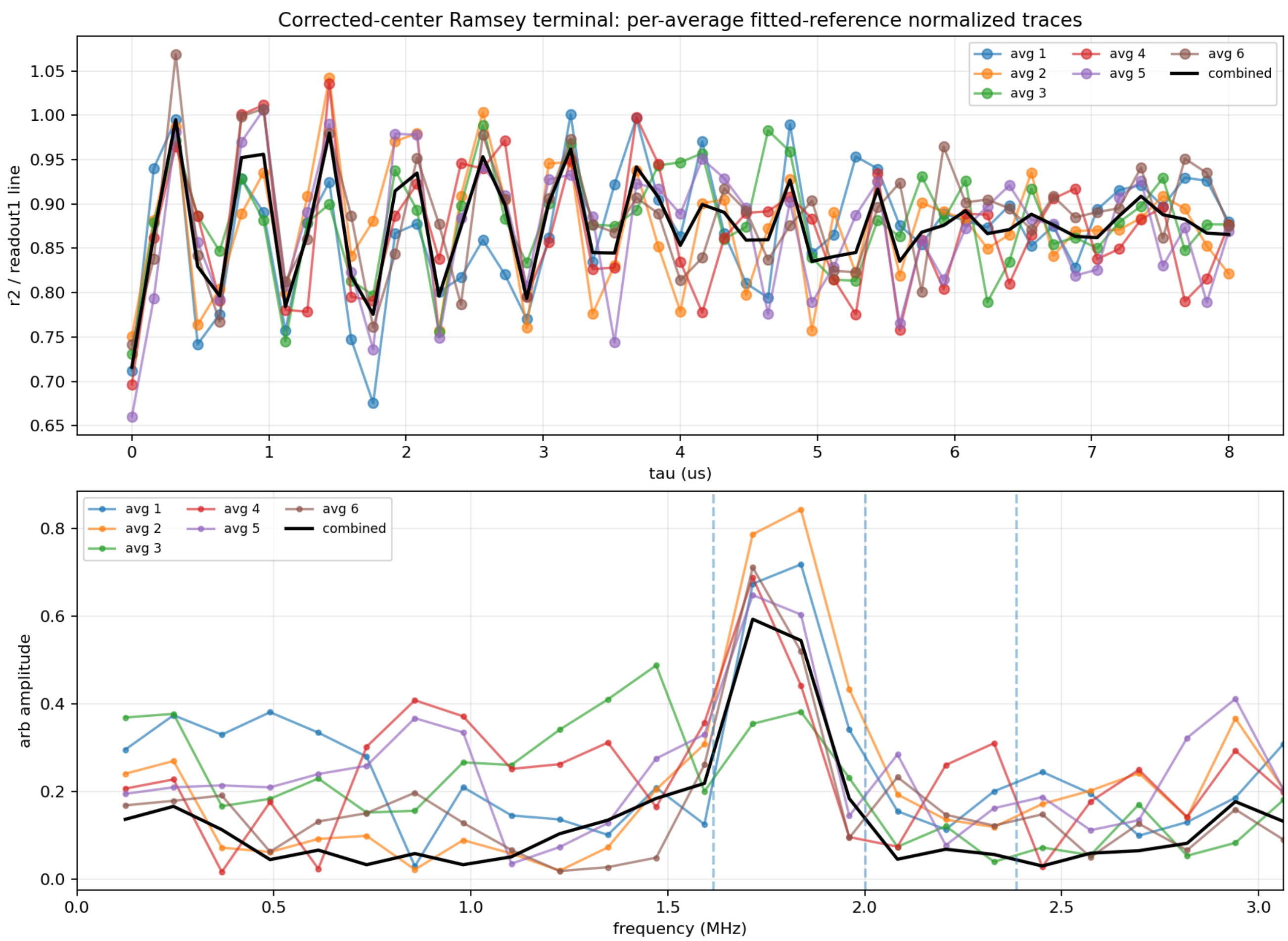

Figure 2: Per-average normalized Ramsey traces and FFTs. The dominant band is visible across stored averages, supporting a real Ramsey signal rather than a one-average artifact.

^{13}C assessment

Using the weak-pi center and the project planning model, the expected ^{13}C Larmor scale was about 384 kHz. For the 2 MHz Ramsey detuning, the expected FFT markers were therefore near 1.616 MHz, 2.000 MHz, and 2.384 MHz. In the terminal fitted-reference-normalized FFT, the carrier target bin ranks 5, the lower ^{13}C target bin ranks 3, and the upper target bin ranks 19.

This is not a claim-grade ^{13}C signature. The lower target bin is present, but it is part of a broad carrier/detuning-like band centered around 1.76–1.84 MHz. The upper target bin is weak, and the data do not show a clean symmetric sideband pair around a resolved carrier. The supported conclusion is therefore: no resolved nearby ^{13}C coupling is observed in the corrected-center Ramsey data. This is a negative claim at the current measurement sensitivity and analysis scope, not a proof that no nuclear spin exists anywhere near the NV.

Drift and data-quality checks

The scan-order-aware drift diagnostic used `Scan.ScanOrderEachAvg` in snake mode for all 6 averages and flagged no averages. Raw readout 1 changed slowly and was handled by fitted-reference normalization; raw readout 2 and normalized views agree on the main oscillatory band. The signal amplitude is large compared with the FFT non-DC median noise floor, but the exact T_2^* value remains model-dependent.

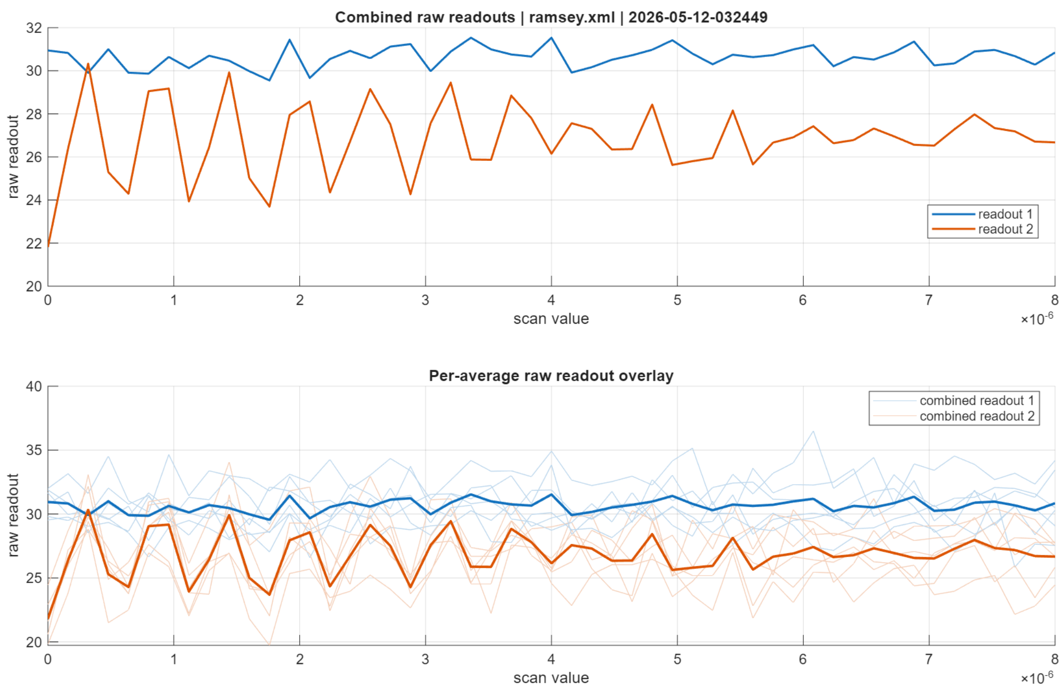

Figure 3: Raw readout export diagnostic for the terminal Ramsey savedexperiment.

Literature and prior-result context

The default NV reference in the local literature library is Doherty et al., "The nitrogen-vacancy colour centre in diamond," Physics Reports 528, 1–45 (2013), DOI 10.1016/j.physrep.2013.02.001. Crossref metadata for this DOI was checked on 2026-05-12. The project used the standard aligned-band planning approximation to estimate the field and ^{13}C Larmor frequency; the exact calculation is recorded in `work/notes/image231924_c01_ramsey_fft_expectation_20260512_0130.md`.

Closeout decision

The human objective is satisfied for `image231924_c01`: a magnetic-field-aligned NV was found, T_2^* has a supported scale, and the ^{13}C check has a supported negative conclusion at the current measurement sensitivity. Further bridge work is optional only if Takuya wants a higher-precision T_2^* estimate or a stronger null limit on weak ^{13}C coupling.